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ECEN 300: PS12

7.27(a), 7.31(a), 7.32(b), 7.35(a)(b), 7.42(c), MATLAB 7.58

7.27 (a) Determine the impulse response corresponding to the following transfer function if:
(i) the system is stable
(ii) the system is causal

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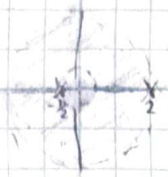
(a) $H(z) = \dfrac{2 - \frac{3}{2}z^{-1}}{(1-2z^{-1})(1+\frac{1}{2}z^{-1})}$

poles: $-\frac{1}{2}, 2$

(i) stable: include unit circle



first: $|z| < 2$
second: $|z| > \frac{1}{2}$
together: $\frac{1}{2} < |z| < 2$

partial frac: $\dfrac{A}{1-2z^{-1}} + \dfrac{B}{1+\frac{1}{2}z^{-1}} = \dfrac{2-\frac{3}{2}z^{-1}}{(1-2z^{-1})(1+\frac{1}{2}z^{-1})} = \underbrace{\dfrac{1}{1-2z^{-1}}}_{|z|<2} + \underbrace{\dfrac{1}{1+\frac{1}{2}z^{-1}}}_{|z|>\frac{1}{2}}$

$A(1+\frac{1}{2}z^{-1}) + B(1-2z^{-1}) = 2 - \frac{3}{2}z^{-1}$
$A(1+\frac{1}{2}x) + B(1-2x) = 2 - \frac{3}{2}x$
$A + \frac{1}{2}Ax + B - 2Bx = 2 - \frac{3}{2}x$
$A + B = 2$
$\frac{1}{2}A - 2B = -\frac{3}{2}$

$\begin{bmatrix} 1 & 1 & 2 \\ \frac{1}{2} & -2 & -\frac{3}{2} \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & \cdot \\ 0 & 1 & \cdot \end{bmatrix}$

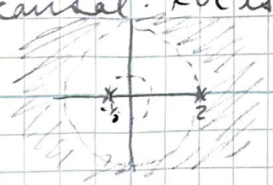
E.1.1: $\mathcal{Z}^{-1}\{-\alpha^n u[-n-1]\} = \dfrac{1}{1-\alpha z^{-1}}$ ROC: $|z| < |\alpha|$

E.1: $\mathcal{Z}^{-1}\{\alpha^n u[n]\} = \dfrac{1}{1-\alpha z^{-1}}$, ROC $= |z| > \alpha$

$$\boxed{h[n] = -2^n u[-n-1] + (-\tfrac{1}{2})^n u[n]}$$

* notecard: transfer is z-transform $H(z)$ of the impulse response $h[n]$

(ii) causal: ROC is expanding



first: $|z| > 2$
second: $|z| > \frac{1}{2}$
together: $|z| > 2$

$$\boxed{h[n] = (2)^n u[n] + (-\tfrac{1}{2})^n u[n]}$$

7.31 (a) Determine (i) transfer function and
(ii) impulse response representations of the causal systems described by the following difference equations.

(a) $y[n] - \frac{1}{2}y[n-1] = 2x[n-1]$

E.2 $\mathcal{Z}\{y[n-k]\} = z^{-k}X(z)$

(i) $Y(z) - \frac{1}{2}z^{-1}Y(z) = 2z^{-1}X(z)$

$Y(z)\left[1 - \frac{1}{2}z^{-1}\right] = 2z^{-1}X(z)$

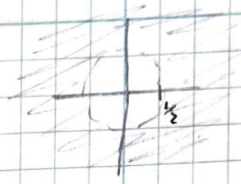
$H(z) = \frac{Y(z)}{X(z)} = \frac{2z^{-1}}{1-\frac{1}{2}z^{-1}}$

$$\boxed{H(z) = \frac{2z^{-1}}{1-\frac{1}{2}z^{-1}}}$$ ← transfer function

(ii) causal : ROC is expanding

poles : $|z| = \frac{1}{2}$

ROC $= |z| > \frac{1}{2}$



$H(z) = 2z^{-1} \cdot \underbrace{\frac{1}{1-\frac{1}{2}z^{-1}}}_{= A(z)}$

$\mathcal{Z}^{-1}\{\ \} = \left(\frac{1}{2}\right)^n u[n] = a[n]$

$\mathcal{Z}^{-1}\{2z^{-1}A(z)\} = 2a[n-1] = 2\left[\left(\frac{1}{2}\right)^{n-1}u[n-1]\right]$

$$\boxed{h[n] = 2\left(\frac{1}{2}\right)^{n-1}u[n-1]}$$ ← impulse response

7.32(b) Determine (i) transfer function & (ii) difference equation from impulse response:

(b) $h[n] = (\frac{1}{3})^n u[n] + (\frac{1}{2})^{n-2} u[n-1]$

E.1: $\mathcal{Z}\{\alpha^n u[n]\} = \frac{1}{1-\alpha z^{-1}}, \; |z|>|\alpha|$

E.2: $\mathcal{Z}\{x[n-k]\} = z^{-k}X(z)$, R_x except possibly $|z| = 0, \infty$

$$h[n] = (\tfrac{1}{3})^n u[n] + (\tfrac{1}{2})^{-1} \cdot (\tfrac{1}{2})^{n-1} u[n-1]$$

$$= \underbrace{(\tfrac{1}{3})^n u[n]}_{a[n]} + 2\underbrace{(\tfrac{1}{2})^{n-1} u[n-1]}_{b[n]} \qquad \underbrace{(\tfrac{1}{2})^n u[n]}_{c[n]}$$

$$A(z) = \frac{1}{1-\frac{1}{3}z^{-1}} \; ; \; C(z) = \frac{1}{1-\frac{1}{2}z^{-1}}$$

$$b[n] = 2c[n-1]$$

$$B(z) = 2z^{-1} \cdot \frac{1}{1-\frac{1}{2}z^{-1}}$$

$$\boxed{H(z) = \frac{1}{1-\frac{1}{3}z^{-1}} + \frac{2z^{-1}}{1-\frac{1}{2}z^{-1}}}$$ transfer function

(ii) $$H(z) = \frac{(1-\frac{1}{2}z^{-1})}{(1-\frac{1}{3}z^{-1})(1-\frac{1}{2}z^{-1})} + \frac{2z^{-1}(1-\frac{1}{3}z^{-1})}{(1-\frac{1}{3}z^{-1})(1-\frac{1}{2}z^{-1})}$$

$$= \frac{1-\frac{1}{2}z^{-1} + 2z^{-1} - \frac{2}{3}z^{-2}}{1-\frac{1}{2}z^{-1}-\frac{1}{3}z^{-1}+\frac{1}{6}z^{-2}} = \frac{1+\frac{3}{2}z^{-1}-\frac{2}{3}z^{-2}}{1+\frac{5}{6}z^{-1}+\frac{1}{6}z^{-2}} = \frac{Y(z)}{X(z)}$$

$$Y(z)\left[1+\tfrac{5}{6}z^{-1}+\tfrac{1}{6}z^{-2}\right] = X(z)\left[1+\tfrac{3}{2}z^{-1}-\tfrac{2}{3}z^{-2}\right]$$

$$\boxed{y[n] - \tfrac{5}{6}y[n-1] + \tfrac{1}{6}y[n-2] = x[n] + \tfrac{3}{2}x[n-1] - \tfrac{2}{3}x[n-2]}$$

difference equation

7.35 (a) For each of the following systems, identify the transfer function of the inverse system, and determine whether the inverse system can be both causal and stable:

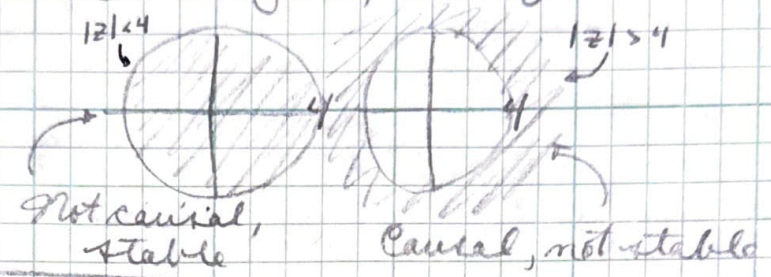
$$(a)\quad H(z) = \frac{1 - 8z^{-1} + 16z^{-2}}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}$$

$$H^{inv}(z) = \frac{1}{H(z)} = \boxed{\frac{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}{1 - 8z^{-1} + 16z^{-2}}}$$

$* z^{-1} = x$ (+ → −)

$$\frac{\frac{1}{4}x^2 - \frac{1}{2}x + 1}{16x^2 - 8x + 1} = \frac{\frac{1}{4}x^2 - \frac{1}{2}x + 1}{16(x - \frac{1}{4})^2} = \frac{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}{16(z^{-1} - \frac{1}{4})^2}$$

poles: $z = 4$ with this pole, two options:



This system can't be both causal & stable.

7.35 (b) Same as above, with:

$$H(z) = \frac{z^2 - \frac{81}{100}}{z^2 - 1}$$

$$\boxed{H^{inv}(z) = \frac{z^2 - 1}{z^2 - \frac{81}{100}}} = \frac{z^2 - 1}{(z - \frac{9}{10})^2} \quad \text{poles: } z = \frac{9}{10}$$

Two options:

It is possible for this system to be both causal and stable, if the ROC is $|z| > \frac{9}{10}$. This ROC includes the unit circle and is expanding.

7.42 (c) Use the unilateral z-transform to determine the forced response, the natural response, and the complete response of the system described by the following difference equation with the given input & initial conditions:

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(c) $y[n] - \frac{1}{4}y[n-1] - \frac{1}{8}y[n-2] = x[n] + x[n-1]$;

$y[-1] = 1$; $y[-2] = -1$; $x[n] = 3^n u[n]$

E.2.2:

$$x[n-k] \xleftrightarrow{\mathcal{Z}_u} x[-k] + x[-k+1]z^{-1} + \dots + x[-1]z^{-k+1} + z^{-k}X(z) \quad \text{For } k>0$$

$$y[n] - \frac{1}{4}\left[y[-1] + z^{-1}Y(z)\right] - \frac{1}{8}\left[y[-2] + y[-1]z^{-1} + z^{-2}Y(z)\right] = \frac{1}{1-3z^{-1}} + \frac{z^{-1}}{1-3z^{-1}}$$

$$Y(z) - \frac{1}{4}\overbrace{y[-1]}^{1} + \frac{1}{4}Y(z)z^{-1} - \frac{1}{8}\overbrace{y[-2]}^{-1} + \frac{1}{8}y[-1]z^{-1} + \frac{1}{8}Y(z)z^{-2} = \frac{1+z^{-1}}{1-3z^{-1}}$$

$$Y(z) - \frac{1}{4} + \frac{1}{4}Y(z)z^{-1} + \frac{1}{8} + \frac{1}{8}z^{-1} + \frac{1}{8}Y(z)z^{-2} = \frac{1+z^{-1}}{1-3z^{-1}}$$

$$Y(z) - \frac{1}{4}Y(z)z^{-1} + \frac{1}{8}Y(z)z^{-1} = \frac{1}{4} - \frac{1}{8} - \frac{1}{8}z^{-1} + \frac{1+z^{-1}}{1-3z^{-1}}$$

$$Y(z)\left[1 - \frac{1}{4}z^{-1} + \frac{1}{8}z^{-2}\right] = \frac{1}{4} - \frac{1}{8} - \frac{1}{8}z^{-1} + \frac{1+z^{-1}}{1-3z^{-1}}$$

$$Y(z) = \frac{\frac{1}{4} - \frac{1}{8} - \frac{1}{8}z^{-1} + \frac{1+z^{-1}}{1-3z^{-1}}}{1 - \frac{1}{4}z^{-1} + \frac{1}{8}z^{-2}} = \frac{-\frac{1}{8} - \frac{1}{8}z^{-1} + \frac{1+z^{-1}}{1-3z^{-1}}}{1 + \frac{1}{4}z^{-1} + \frac{1}{8}z^{-2}}$$

$$= \frac{-\frac{1}{8} - \frac{1}{8}z^{-1} + \frac{1+z^{-1}}{1-3z^{-1}}}{(1-\frac{1}{2}z^{-1})(1+\frac{1}{4}z^{-1})} = \frac{-\frac{1}{8}}{(1-\frac{1}{2}z^{-1})(1+\frac{1}{4}z^{-1})} + \frac{-\frac{1}{8}z^{-1}}{(1-\frac{1}{2}z^{-1})(1+\frac{1}{4}z^{-1})} + \frac{1+z^{-1}}{(1-3z^{-1})}$$

$$\frac{A}{1-\frac{1}{2}z^{-1}} + \frac{B}{1+\frac{1}{4}z^{-1}} = \frac{-\frac{1}{8} - \frac{1}{8}z^{-1}}{(1-\frac{1}{2}z^{-1})(1+\frac{1}{4}z^{-1})}$$

$$A(1+\tfrac{1}{4}z^{-1}) + B(1-\tfrac{1}{2}z^{-1}) = -\tfrac{1}{8} - \tfrac{1}{8}z^{-1}$$

$$A + \tfrac{1}{4}Az^{-1} + B - \tfrac{1}{2}Bz^{-1} = -\tfrac{1}{8} - \tfrac{1}{8}z^{-1}$$

$$A + B = -\tfrac{1}{8}$$

$$\tfrac{1}{4}A - \tfrac{1}{2}B = -\tfrac{1}{8}$$

$$\begin{bmatrix} 1 & 1 & -\frac{1}{8} \\ \frac{1}{4} & -\frac{1}{2} & -\frac{1}{8} \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & -\frac{1}{4} \\ 0 & 1 & \frac{1}{8} \end{bmatrix}$$

$$\frac{A}{1-\frac{1}{2}z^{-1}} + \frac{B}{1+\frac{1}{4}z^{-1}} + \frac{C}{1-3z^{-1}} = \frac{1+z^{-1}}{(\ )(\ )}$$

$$A(1+\tfrac{1}{4}z^{-1})(1-3z^{-1}) + B(1-\tfrac{1}{2}z^{-1})(1-3z^{-1}) + C(1-\tfrac{1}{2}z^{-1})(1+\tfrac{1}{4}z^{-1}) = 1+z^{-1}$$

$$A(1 + -\tfrac{1}{4}z^{-1} - \tfrac{3}{4}z^{-2})$$

$$B(1 - \tfrac{7}{2}z^{-1} + \tfrac{3}{2}z^{-2})$$

$$C(1 - \tfrac{1}{4}z^{-1} - \tfrac{1}{8}z^{-2}) = 1 + z^{-1}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ -\frac{1}{4} & -\frac{7}{2} & -\frac{1}{4} & 1 \\ -\frac{3}{4} & \frac{3}{2} & -\frac{1}{8} & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} a & -2/5 \\ a & -1/13 \\ a & 96/65 \end{bmatrix}$$

$$Y(z) = \frac{-1/4}{1-\frac{1}{2}z^{-1}} + \frac{1/8}{1+\frac{1}{4}z^{-1}} + \frac{96/65}{1-3z^{-1}} - \frac{2/5}{1-\frac{1}{2}z^{-1}} - \frac{1/13}{1+\frac{1}{4}z^{-1}}$$

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$$\boxed{y[n] = \underbrace{-\frac{1}{4}\left(\frac{1}{2}\right)^n u[n] + \frac{1}{8}\left(-\frac{1}{4}\right)^n u[n]}_{\text{forced response}} + \underbrace{\frac{96}{65}(3)^n u[n] - \frac{2}{5}\left(\frac{1}{2}\right)^n u[n] - \frac{1}{13}\left(-\frac{1}{4}\right)^n u[n]}_{\text{natural response}}}$$

complete response